

Supervised Learning

CORE MODELS



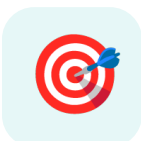
Linear Regression

Predicts continuous numbers with roughly linear relationships.



Logistic Regression

Binary classification using a threshold (e.g., Spam Yes/No).



K-Nearest Neighbors

Predicts based on local similarity of K closest points.

⚠️ **Requires Feature Scaling**



Support Vector Machine

Finds optimal hyperplane for clear margins in high-dim data.

⚠️ **Requires Feature Scaling**



Naive Bayes

Fast baseline for text classification and spam filtering. Assumes feature independence.

TREE-BASED MODELS



Decision Trees

Splits data recursively. Highly interpretable rules but prone to overfitting.



Random Forest

Robust ensemble of random trees. Excellent baseline for tabular data.



Boosting XGBoost

Builds trees sequentially to correct prior errors. High accuracy for structured data competitions.

Advanced Paradigms

UNSUPERVISED LEARNING



K-Means

Groups data into K clusters. Very sensitive to outliers.



DBSCAN

Density-based clustering. Automatically finds outliers/noise.



PCA

Dimensionality reduction technique projecting maximum variance.



Association Rules (Apriori)

Identifies frequent itemsets and 'if-then' rules. Ideal for market basket analysis.

DEEP & REINFORCEMENT LEARNING

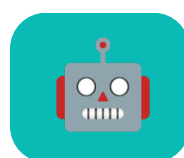


Deep Learning

TensorFlow

PyTorch

- **CNNs:** Spatial patterns & images.
- **Transformers:** State-of-the-art for sequences/text.
- **RNN/LSTM:** Standard sequential processing.



Reinforcement Learning

Agent learns by interacting with an environment, using **rewards and penalties** to optimize decisions over time.

Algorithm Selection

WHAT IS YOUR GOAL?

Predict a Number	→	Linear Reg (Simple) OR Random Forest / XGBoost
Classify Categories	→	Logistic Reg (Explainable) OR SVM / Ensembles
Group Data	→	K-Means (Known Groups) OR DBSCAN
Reduce Features / Visualize	→	PCA (Dimensions)
Find Relationships	→	Apriori / Association Rules
Images or Text	→	CNNs (Visual) OR Transformers / Naive Bayes (Text)
Optimize Decisions	→	Reinforcement Learning (RL)